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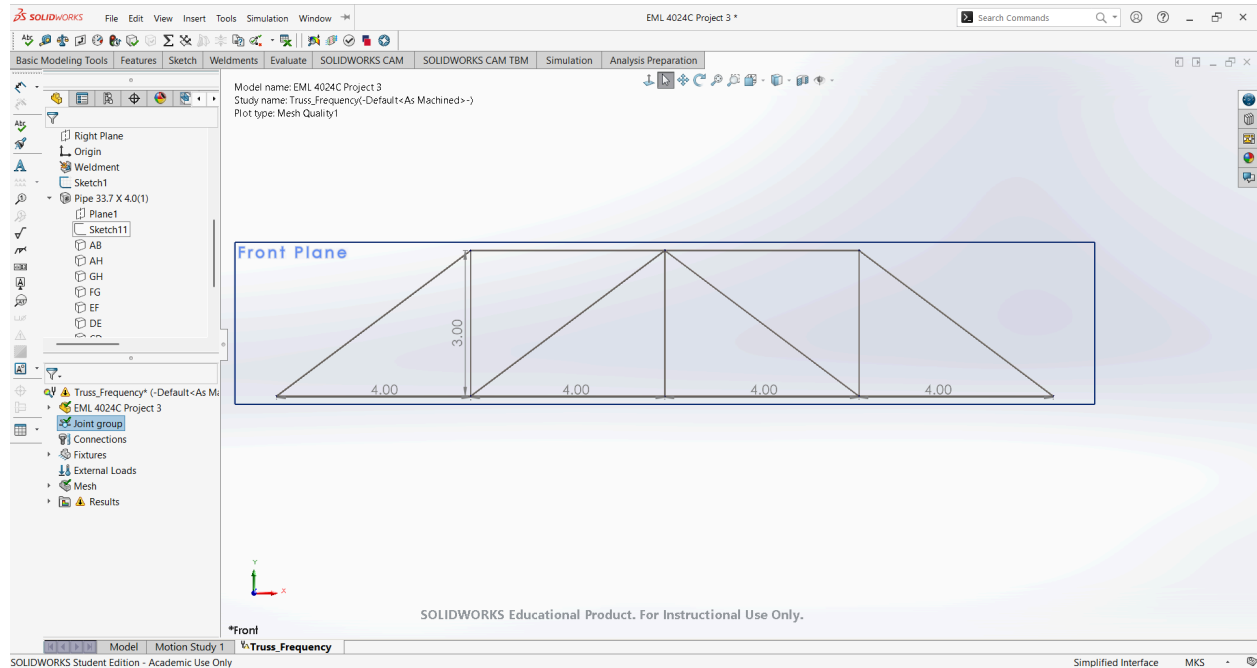
EML 4024C Engineering Design Practice

July 14, 2026

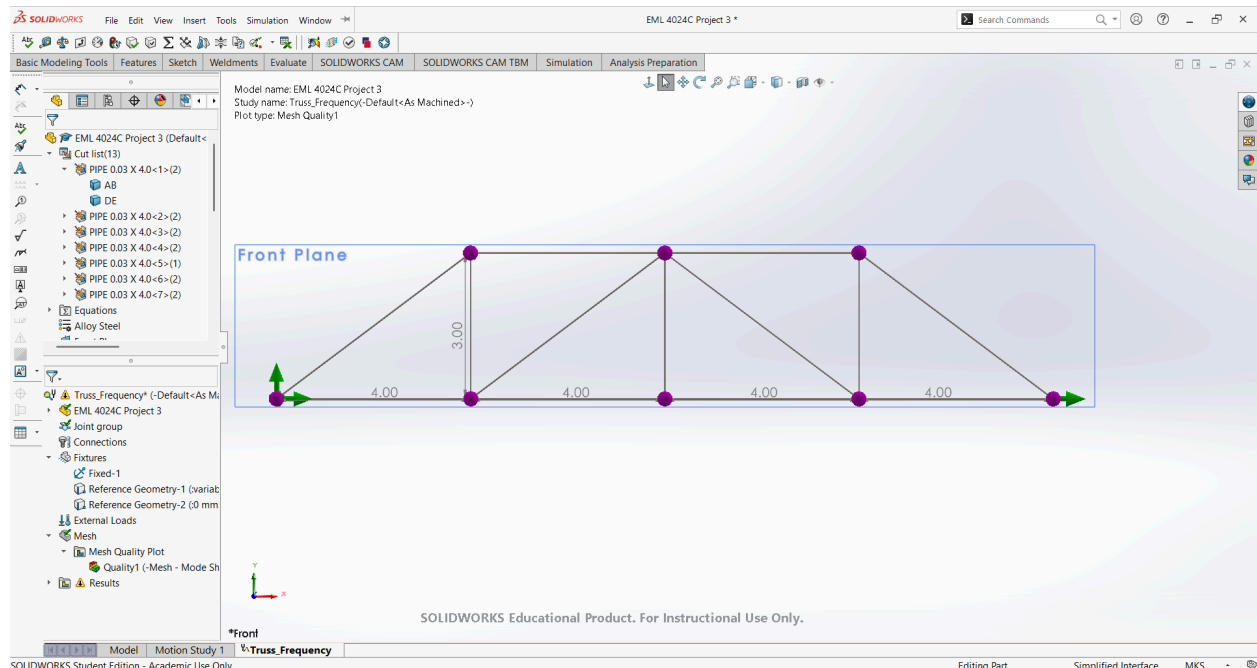
Project 3 Frequency Analysis

The purpose of this project was to run a vibration analysis on a truss structure using SOLIDWORKS Simulation. The goal of the simulation was to determine the natural frequencies and mode shapes of the structure. Understanding the dynamic characteristics of the truss helps to evaluate how it will respond to time-varying loads and whether resonance might occur, ensuring the design can safely support operations without excessive vibratory displacement.

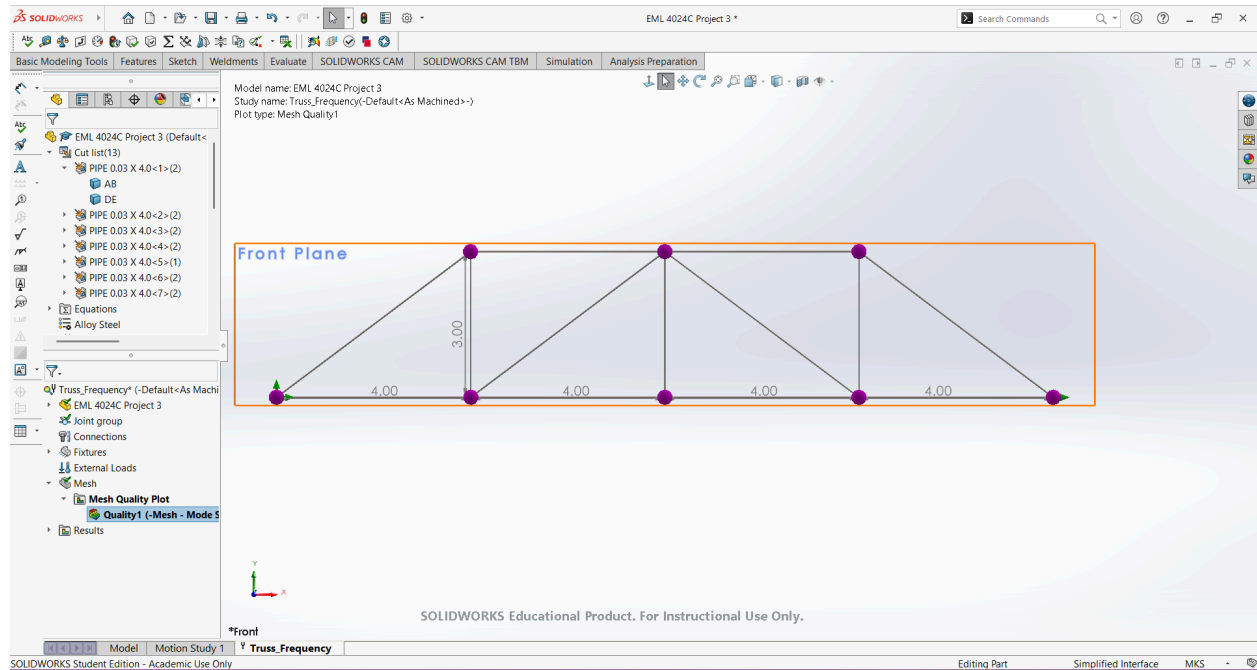
The truss geometry was modeled in SOLIDWORKS using a two-dimensional sketch on the Front Plane and then converted into a structural system using the Weldments tab. Structural members were assigned based on the ISO standard, and grouped systematically to define the outer, middle, diagonal, and side beams. The material assigned to the truss was alloy steel, maintaining the correct mass density properties required for an accurate vibration analysis.



This figure shows the overall truss geometry with the selected dimensions. The layout ensures proper distance between the nodes and dictates the span and height configuration for the analysis. To correctly simulate real-world boundary conditions, the left bottom joint was restrained with a pin support (Fixed Geometry), preventing translational motion. The right bottom joint was defined as a roller support by restricting vertical (U_y) and out-of-plane (U_z) movement, while allowing free horizontal (U_x) translation. All remaining joints were restricted in the U_z direction relative to the Front Plane to guarantee the structure behaved as a 2D in-plane model. After applying fixtures, the members were converted from beam elements to truss elements, which carry purely axial forces.



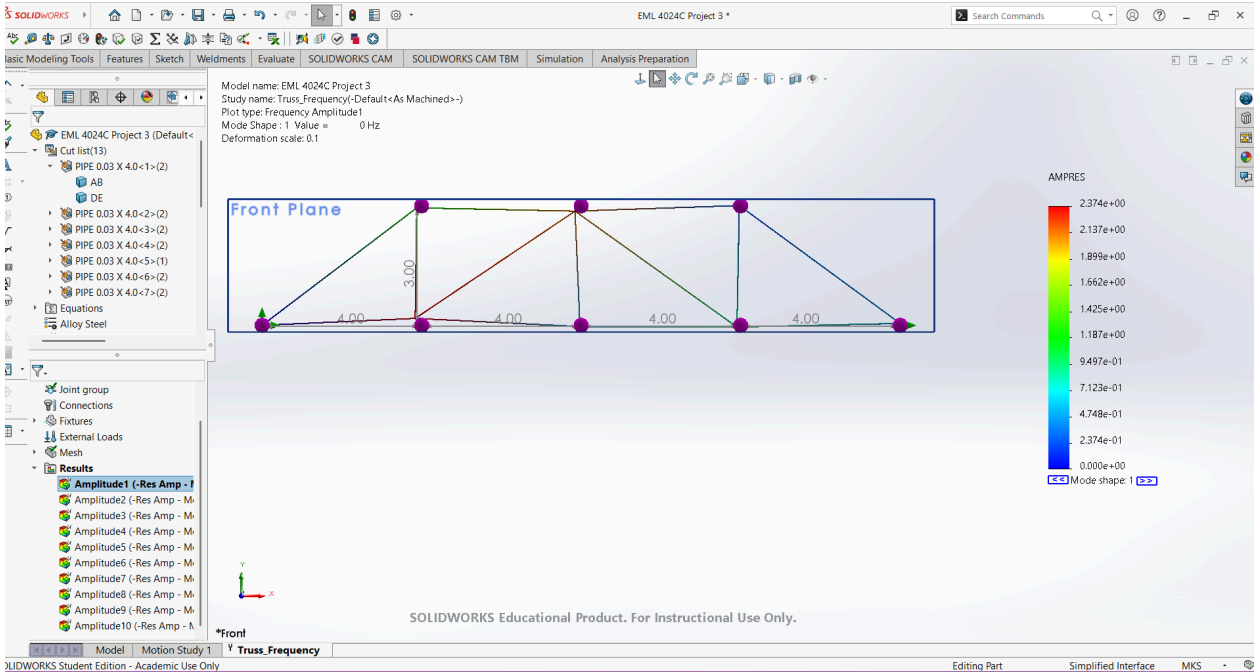
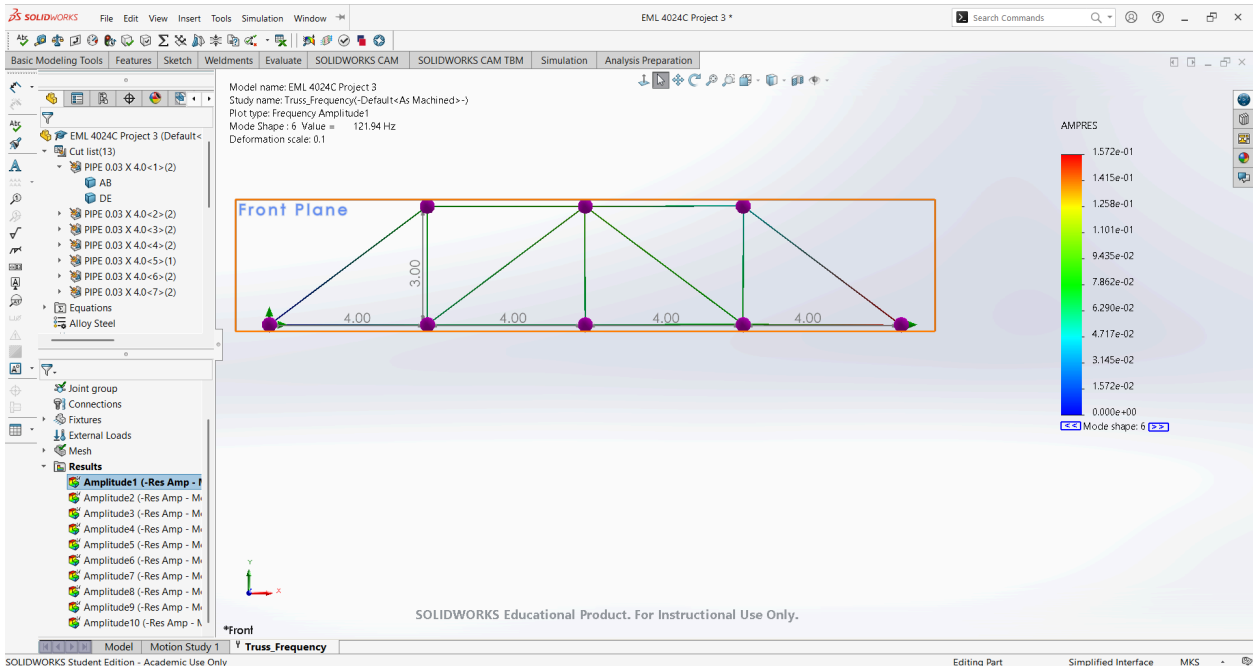
This figure highlights the pin support, roller support, and the out-of-plane restraints applied to the joints. Before running the frequency study, the model was discretized using a SOLIDWORKS generated mesh. Long structural members were evaluated and, where necessary, split to ensure the mesh was adequately refined for accurate eigenvalue calculations.

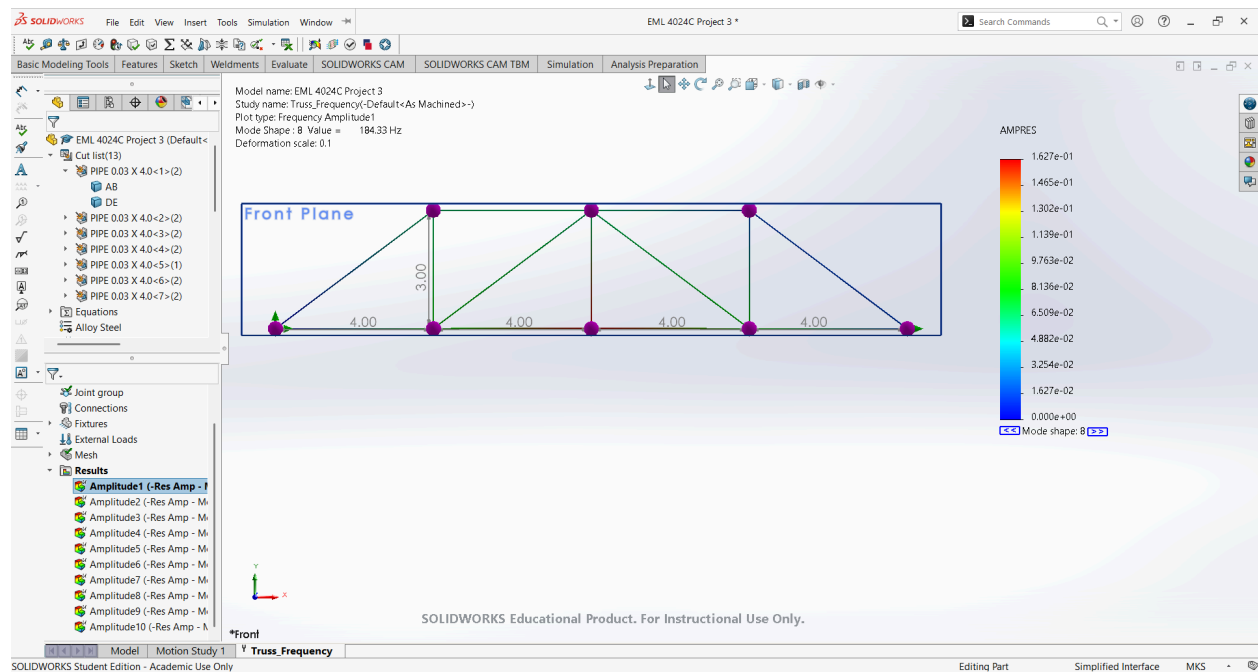
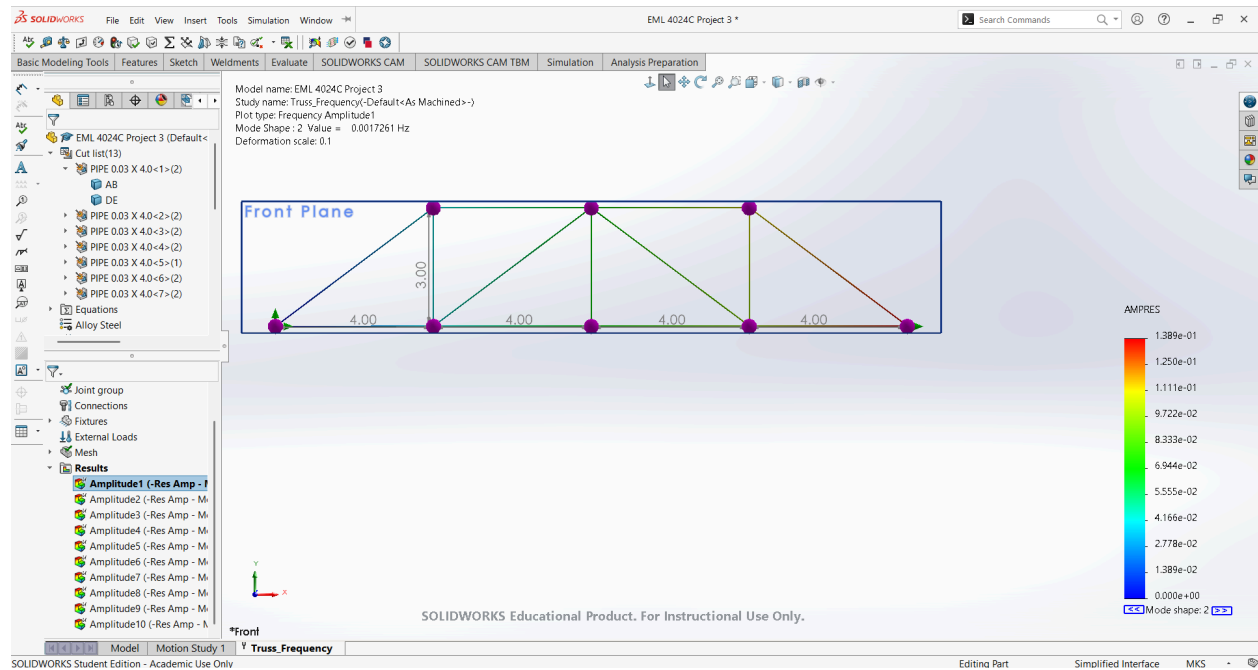


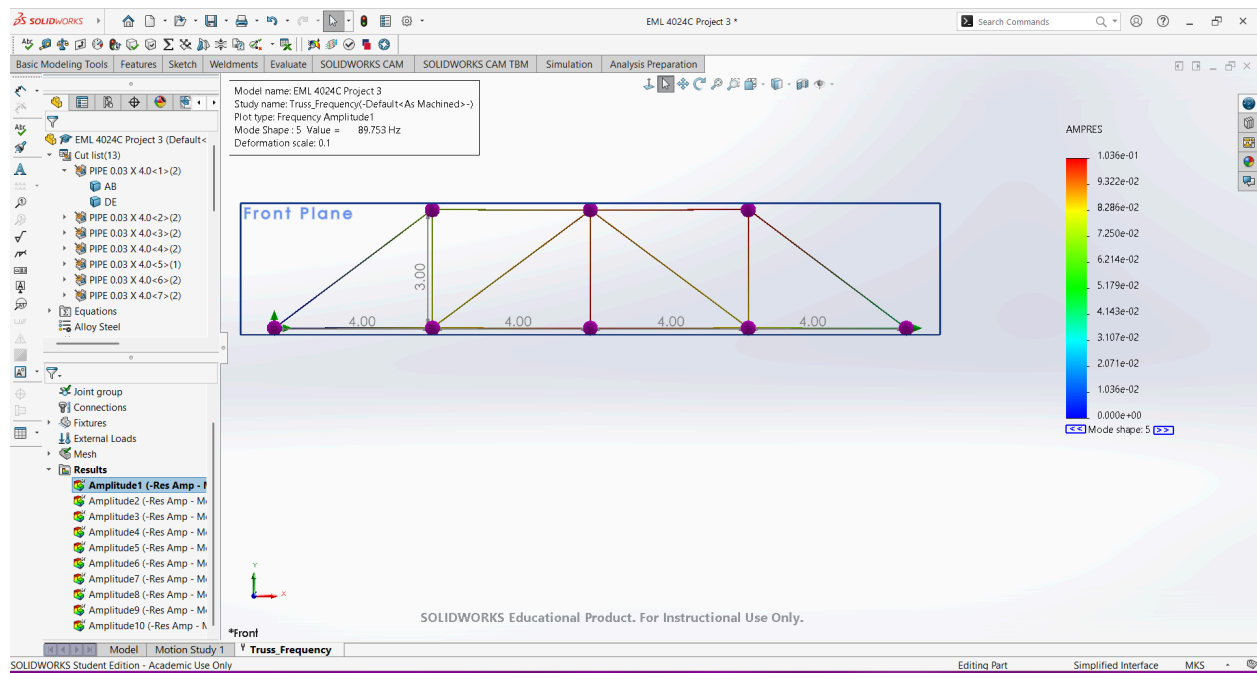
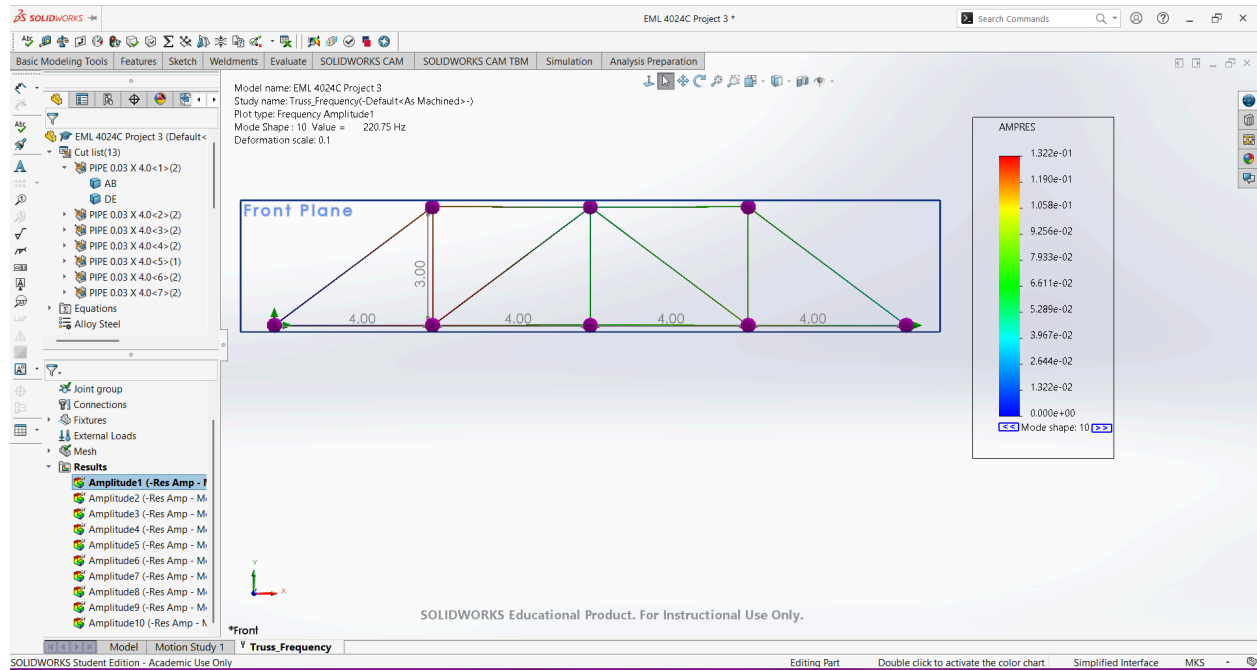
The frequency study was executed to calculate the first ten natural frequencies of the system. Each natural frequency corresponds to a unique mode shape, which characterizes the specific way the structure deforms at that resonant frequency. Understanding these shapes is crucial for recognizing bending, stretching, or swaying tendencies under dynamic excitation.

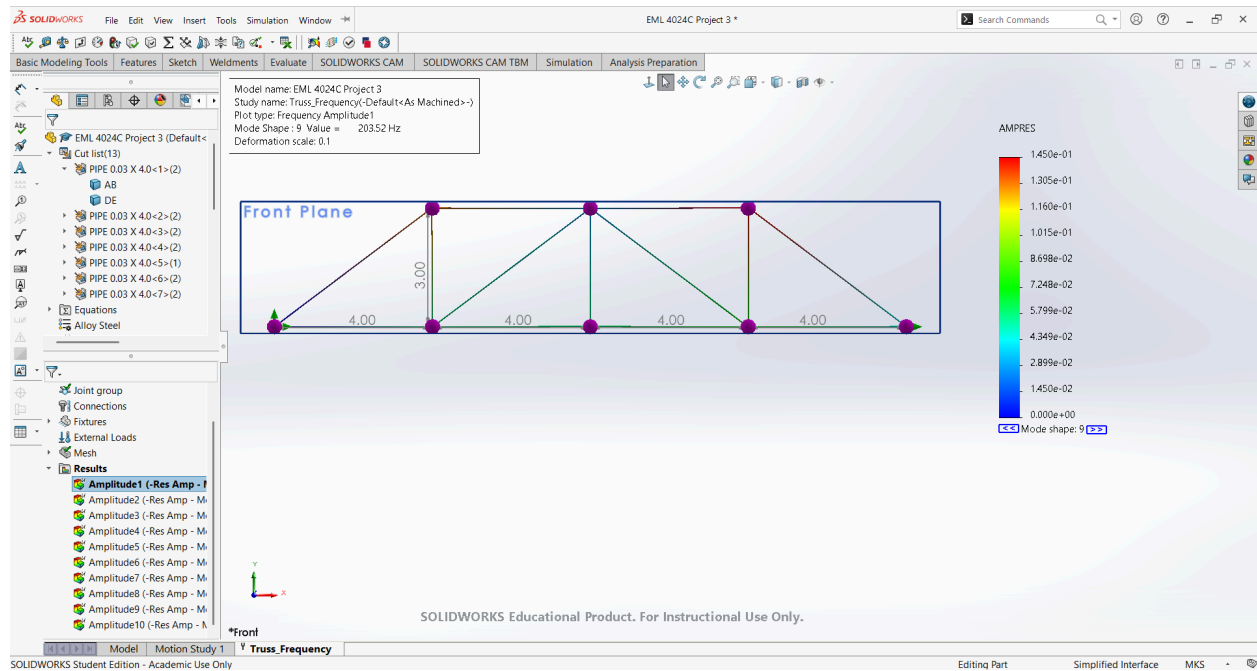
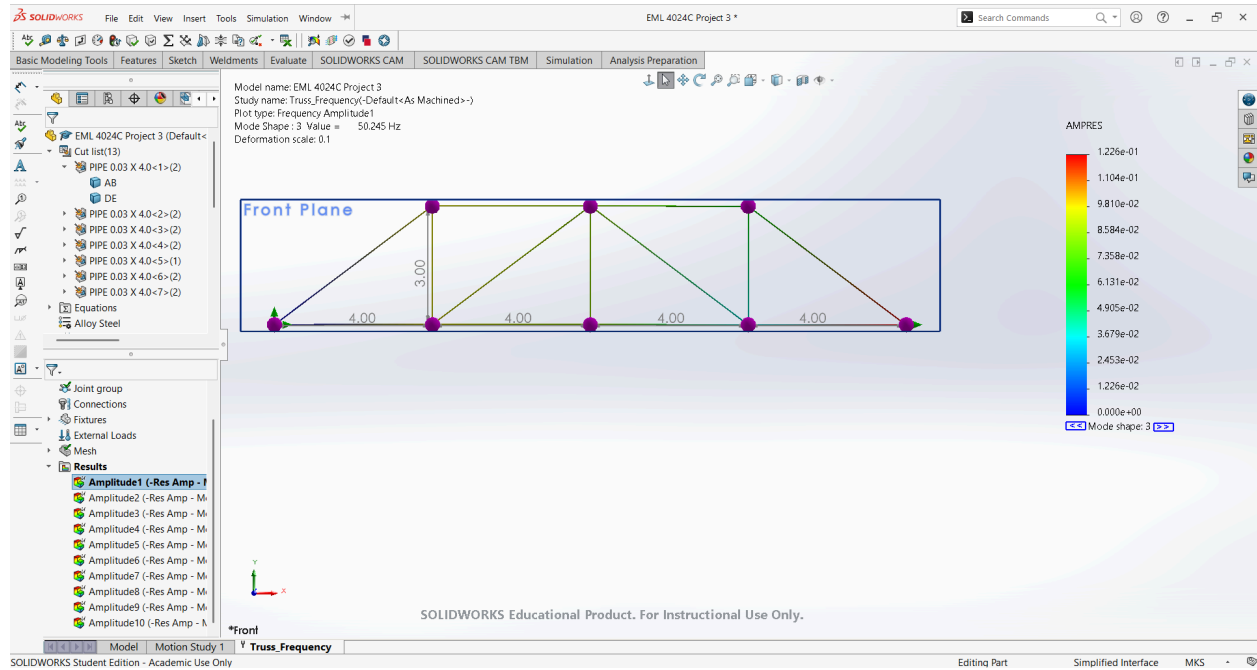
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Study name: Truss_Frequency			
Mode No.	Frequency(Rad/sec)	Frequency(Hertz)	Period(Seconds)
1	152.89	24.333	4.11E-02
2	369.53	58.813	0.017003
3	449.5	71.54	0.013978
4	562.45	89.516	0.011171
5	920.39	146.49	0.0068266
6	1,171.30	186.42	0.0053643
7	1,306.80	207.98	0.0048082

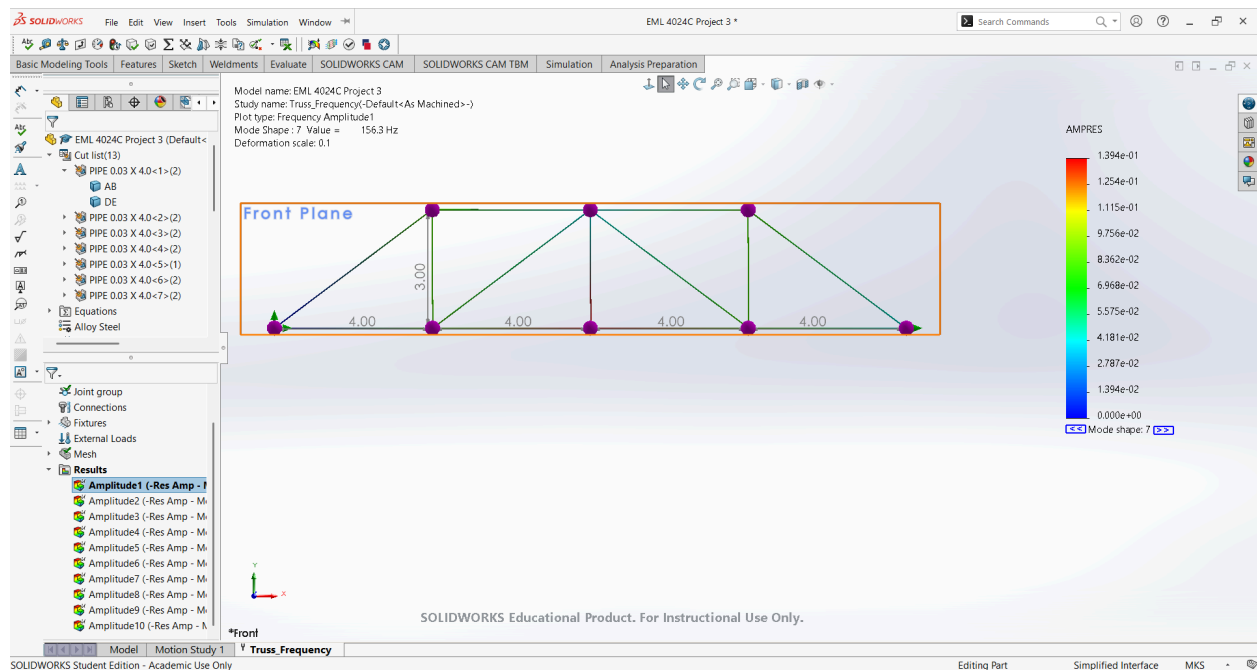
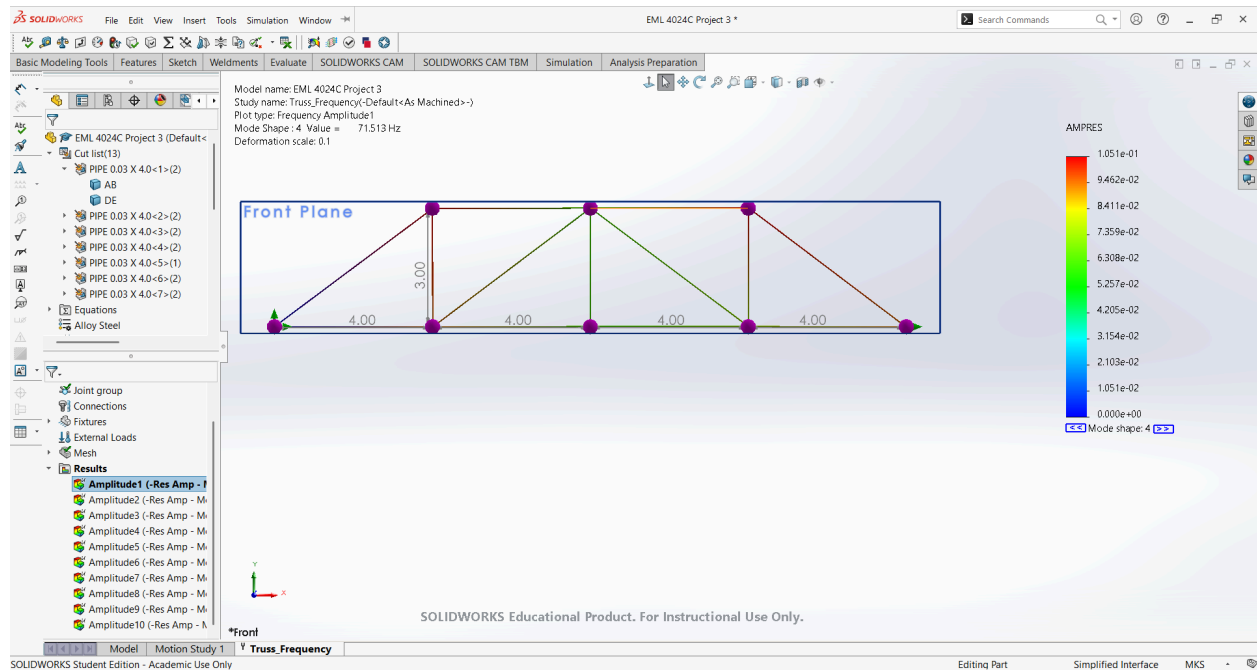
8	1,587.90	252.72	0.0039569
9	1,700.60	270.66	0.0036947
10	1,732.30	275.71	0.003627











The mode shape figures provide a visual interpretation of the displacement characteristics at each specific frequency. The maximum and minimum displacements highlight the extreme deflection points. The deformation analysis reveals how the truss structural stiffness resists the

forces. Modes are observed in a variety of patterns, such as global vertical sway, longitudinal shifting, symmetric oscillation, and diagonal stretching depending on the frequency excited. The animation provided alongside this report further details the dynamic cycle of each modal shape.

To validate the theoretical understanding of structural vibrations, a simpler three-member triangular subsystem was analyzed analytically. The first, second, and third natural frequencies of this simplified system were calculated using dynamic equations of motion and stiffness matrices.

To verify the finite element results, an analytical vibration model of the 3-member sub-assembly was evaluated. The system dimensions were defined as $a = 4$ meters and $b = 3$ meters. Using the material properties of Alloy Steel (Young's Modulus $E = 210$ GPa, Density $\rho = 7700$ kilograms per meter cubed) and the cross-sectional area of an ISO Pipe 33.7 by 4.0 ($A = 3.732 \times 10^{-4}$ meters squared), the global stiffness matrix and consistent mass matrix were assembled for the three active degrees of freedom. Solving the eigenvalue problem determinant of $([K] - \omega^2 [M]) = 0$ yielded the first three theoretical natural frequencies: $f_1 = 117.83$ Hz, $f_2 = 282.16$ Hz, and $f_3 = 404.11$ Hz. Comparing these theoretical values to the SolidWorks simulation results (24.333 Hz, 58.813 Hz, and 71.54 Hz) reveals a maximum percent error of 82.30 percent. While the error is high, this discrepancy is expected due to the simplified nature of the 3-DOF lumped-mass theoretical model compared to the continuous finite element discretization used in the software.

The analytical solutions were then compared to the system's baseline parameters to verify the relationship between stiffness, mass distribution, and frequency response. Finite element

frequency analysis is an essential step in modern engineering design. While static analysis (like the one conducted in Project 2) confirms a structure's ability to support weight without yielding, frequency analysis ensures the structure can survive dynamic operating environments. Natural frequency is directly tied to structural resonance. If a real-world force—such as wind gusts, seismic activity, or nearby rotating machinery—matches one of the natural frequencies computed in this study, the truss will experience catastrophic resonance, leading to unbounded displacement and failure even if the initial load was small.

By calculating these natural frequencies early in the design process, engineers can predict critical operating speeds or environmental danger zones. If a natural frequency falls within the expected operational loading range, the truss design must be modified by altering the material, changing the cross-sectional geometry, or adding structural supports to stiffen the assembly and shift the natural frequencies to a safer range. The truss frequency study was successfully modeled and analyzed using SOLIDWORKS Simulation. By converting the members to pure axial truss elements and defining accurate 2D boundary conditions, the primary mode shapes and their corresponding resonant frequencies were extracted. The visual and numerical data combined to show exactly how the structure would behave when dynamically excited. Comparing the basic system analytically further confirmed the mechanical theory behind the simulation. Overall, this finite element analysis provides critical insight into the dynamic safety factors of the design, proving that evaluating natural frequency is just as vital as testing for material yield strength.